Business Process Explanation

PitchGuard: Daily Training Load Recording and Injury-Risk Alert

SHEMA Mfizi Cedrick, ID 29740 | INSY 8311, AUCA | Instructor: Eric Maniraguha

1. How the process starts and progresses

The process starts when a training session ends (S1). The player submits the session effort, as RPE and minutes (A1), and a daily wellness report (A2). If the player has pain or an injury (G1), she also reports it with its location and severity (A3). PitchGuard then validates the entry against the PLAYER table (A4). A valid entry is checked against INJURY for a recent major injury (A6). If there is none, the entry is saved (A8), the acute:chronic load ratio is calculated from SESSION_LOAD (A9), and the risk status is decided (G4). A RED or AMBER status creates a RISK_ALERT (A10). The dashboard is refreshed (A11), the coach reviews the alerts (A12), and the decision is recorded (A14). The process ends there, or ends earlier when an entry is rejected (E1).

2. Actors and their roles

Actor	Role in the process
Player	Submits session effort, wellness and injury reports. Corrects and resubmits if an entry is rejected.
PitchGuard (Oracle PL/SQL)	Validates entries, applies the safety rules, saves data, calculates the load ratio, creates alerts and writes the audit log.
Power BI dashboard	External system connected to Oracle. Shows the RED, AMBER and GREEN player list. It only displays data; it does not change it.
Coach	Reviews alerts, decides whether to rest a player or keep training, and records the decision.

3. Decision points

- **Pain or injury today? (G1):** only players who report pain or an injury also submit an injury report.
- **Valid? (G2):** an invalid entry (unknown player, impossible load value, duplicate) is rejected and logged in AUDIT_LOG (A5).
- **Recent major injury? (G3):** if yes, the load entry is blocked and an alert is raised (A7). An injured player cannot be logged as training.
- **Risk status? (G4):** RED or AMBER creates a RISK_ALERT; GREEN needs no alert.
- **Alert level? (G5):** RED means rest the player or reduce the load; AMBER means keep training and monitor.

4. Information flow

Data moves from the player to PitchGuard as session, wellness and injury data. PitchGuard stores it in SESSION_LOAD, WELLNESS_REPORT and INJURY, and writes every action and every denied attempt to AUDIT_LOG. The load ratio produces RISK_ALERT rows. The dashboard reads RISK_ALERT, PLAYER and SESSION_LOAD and passes a red, amber and green list to the coach. The coach's decision returns to PitchGuard and updates RISK_ALERT. At season end a timer event archives resolved alerts (A15).

5. How it supports management decisions

Today a coach often learns that a player is overworked only after she is injured. In PitchGuard the coach sees each player's risk status before the next session and can reduce her load early. The block rule stops an injured player from being logged as training, and the audit log gives the staff a record of who did what and when. The decision recorded in A14 also shows how past alerts were handled, which helps the club improve its load management over time.